

INTERNET OF THINGS

HOMEWORK

EXERCISE 1

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Forklift Monitoring System

This project is focused on building a smart, affordable system that helps keep track of electric forklifts used in a warehouse environment. The warehouse is divided into two main zones: a small indoor area located underground and a much larger outdoor yard. The main goal is to always know where each forklift is, measure how far it travels each day, calculate its speed, and detect if it experiences any sudden impacts or crashes.

To achieve this, the system uses sensors such as GPS for tracking forklifts outdoors, UWB (Ultra-Wideband) for precise indoor location, and an IMU (Inertial Measurement Unit) to detect motion and impacts. Each forklift has a small device called ESP32-S3 that collects data from all these sensors. When something important happens, such as an impact or movement beyond a certain distance, the data is sent wirelessly using LoRa or Wi-Fi.



Hardware Installed on Each Forklift

Every forklift is equipped with a compact IoT system made of small electronic parts. These parts help the forklift detect motion, sense its position, and send important information wirelessly. All the components are connected to a central controller, which is small, efficient, and capable of handling all operations automatically. This setup is designed to be easy to install and affordable, making it suitable for use on many forklifts across the warehouse.

Sensors

We use a variety of sensors to collect valuable information from each forklift. A GPS sensor is used to determine the forklift's position when it is outdoors in the yard. For indoor environments, where GPS signals cannot reach, we use a UWB sensor, which provides very accurate indoor positioning. An IMU sensor helps detect how the forklift is moving, and it can also identify if there has been an impact or sudden change in motion. A wheel sensor is used to measure how much the wheels have rotated, which helps in calculating both speed and distance. Finally, a small BLE receiver is used to detect when the forklift is near a docking station or a known indoor location.

Processing Unit

The main controller used on each forklift is the ESP32-S3 microcontroller. It is a powerful but low-cost chip that collects all the data from the sensors and manages the operation of the system. Instead, the controller works with simple rules — for example, if a strong impact is detected or if the forklift travels a certain distance, the controller prepares a message and sends the data. This method is simple, reliable, and helps reduce the system's power consumption. The ESP32-S3 also supports Wi-Fi and Bluetooth, which makes it flexible for use in both indoor and outdoor areas.

Communication Module

To send the collected data wirelessly, the system uses a LoRa module, which is excellent for long-range, low-power communication. This is ideal for the large outdoor area where forklifts move freely across long distances. When the forklifts are inside the building, the system switches to using Wi-Fi or Bluetooth if available. This flexible communication setup ensures that the forklifts are always connected, no matter where they are in the warehouse or yard.

Connectivity Strategy, Communication Protocol, and Data Frequency

The system uses a hybrid communication strategy, combining LoRa for outdoor use and Wi-Fi or Bluetooth for indoor use. The protocol used to transmit data is MQTT, which is lightweight and specially designed for small devices like the ESP32-S3. The forklift sends data in two ways:

- First, it sends data when something important happens, such as a sudden impact or when the forklift moves a long distance (event-based).
- Second, it sends data regularly every 5 minutes even if nothing special happened, just to keep the system updated (time-based).

This method helps save battery power and keeps the communication network efficient and uncluttered.

Architecture for Data Collected, Stored, and Shown

The data from the forklifts is first sent to a gateway, which collects all incoming messages and passes them on. The messages go to an MQTT broker, which receives and organizes the data from all the forklifts. Then, a backend program processes this data, cleans it up, and stores it in a database for later use. Finally, a dashboard application displays the data in a user-friendly format. On the dashboard, warehouse managers can see live maps showing the position of each forklift, graphs showing their speed and movement over time, and alerts for any impacts or unexpected behavior. This setup makes it easy to track and manage the forklifts in real-time.

Pseudocode for Logic Running on Each Forklift

```
# Initialize all sensors and communication
```

```
initialize_IMU()
```

```
initialize_GPS()
```

```

initialize_UWB()

initialize_wheel_encoder()

initialize_BLE_receiver()

initialize_MQTT()

set_timer_interval(10 seconds)


# Constants

MOVEMENT_THRESHOLD = 10 meters

IMPACT_THRESHOLD = 2.0 g # Acceleration threshold

TIME_UPDATE_INTERVAL = 5 * 60 # 5 minutes in seconds


# Variables

last_transmit_time = get_current_time()

last_distance = 0

# Main Loop

loop forever:

    wait_for_next_timer_tick()

    Read sensor data

    imu_data = read_IMU()

    gps_data = read_GPS()

    uwb_data = read_UWB()

    wheel_rotation = read_wheel_encoder()

    ble_near_dock = check_BLE()

    Calculate derived values

    current_time = get_current_time()

    current_distance = calculate_distance(wheel_rotation)

    delta_distance = current_distance - last_distance

    last_distance = current_distance

    speed = calculate_speed(wheel_rotation)

    Check conditions for sending data

    impact_detected = imu_data.acceleration > IMPACT_THRESHOLD

    movement_detected = delta_distance > MOVEMENT_THRESHOLD

    time_elapsed = (current_time - last_transmit_time) >= TIME_UPDATE_INTERVAL

    if impact_detected or movement_detected or time_elapsed:

        Choose position source

        if ble_near_dock or gps_data.invalid:

            position = uwb_data

        else:

            position = gps_data

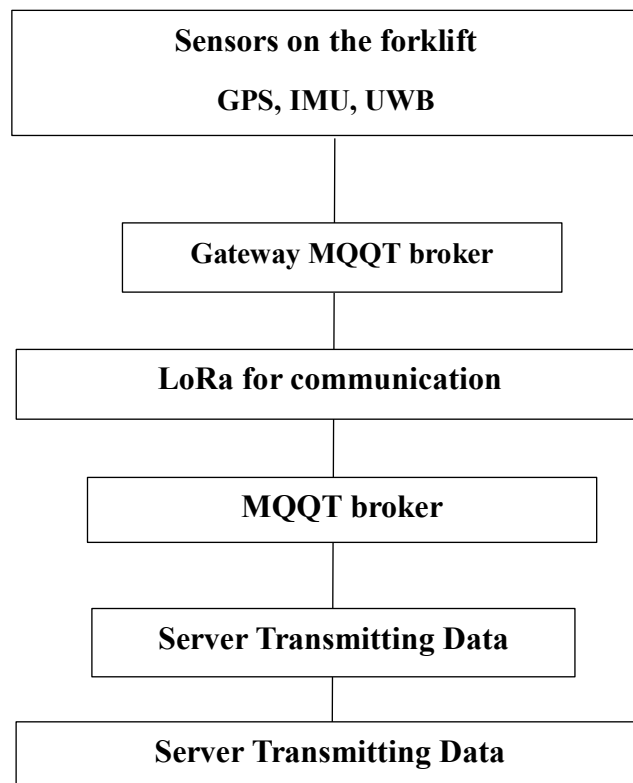
```

Create message payload

```
message = {  
    "timestamp": current_time,  
    "position": position,  
    "impact": impact_detected,  
    "distance": current_distance,  
    "speed": speed  
}
```

Send data

```
mqtt_publish("forklift/telemetry", message)  
last_transmit_time = current_time
```



1. Sensors on the forklift collect movement and location data.
2. The ESP32-S3 microcontroller reads the sensor data and sends it through LoRa, Wi-Fi, or Bluetooth.
3. The data is received by a gateway, which forwards it to an MQTT broker.
4. A backend server processes the data and stores it in a database.
5. A dashboard application displays the forklift's data on maps and graphs in real-time.